Collimator VCurve — Quick Manual (v5.4)

Objective flange focal distance (FFD) measurement for double-pass autocollimator benches. A USB camera replaces your eye at the collimator: the app measures image sharpness (Tenengrad metric) in the R, G and B channels while you step the lens through focus, then fits a parabola to each channel's curve and reports the vertex — the true best-focus position — with its uncertainty. **The G (green) channel is the reference measurement; R–G and B–G show the lens's secondary spectrum.**

Works the same on Windows, macOS and Linux. Touchscreen-friendly.

1. Install & first run

OS	Steps
Windows	Run `CollimatorVCurve-setup.exe` and follow the wizard (no admin needed). Launch from the Start Menu. If SmartScreen appears: *More info → Run anyway*.
macOS	Open the `.dmg`, drag the app to Applications. First launch: **right-click** → **Open** (Gatekeeper). Allow camera access when asked.
Linux	Ubuntu/Debian: install `CollimatorVCurve-amd64.deb` (double-click or `sudo apt install ./CollimatorVCurve-amd64.deb`), launch from the menu. Other distros: `chmod +x CollimatorVCurve-linux-x86_64`, then run it.

Connect the USB camera **before** starting the app. The app opens as a maximized window filling the screen, finds the camera automatically and remembers your last camera, language (PT/EN) and unit (mm/μm) between sessions.

2. The screen

- **Live view** (left): the camera image in the selected channel's color. Drag a rectangle to set the **ROI** — keep it on the reticle image. The sparkline at the bottom shows the live focus metric with its **RMS** readout and a dashed **peak-hold** line marking the best value seen.
- **Zoom** (view only — the measurement always uses the raw image): pinch (= Ctrl + wheel), the **– / +** corner buttons, `+` / `-` keys, or **double-tap** for max zoom. Centered on the ROI; step, max (5×/10×) and gestures configurable in SETTINGS.
- **Side panel** (right): POINT · FIT · CHANNEL · CAM · CLEAR · SETTINGS · PHOTO, plus — at the bottom — the **?** (help) button, the **CLOSE** button and the PT/EN language toggle.
- **CLOSE (bottom of the panel)**: tap twice to quit (first tap asks "QUIT?"). ESC or `q` also work.
- **? (help)**: features, shortcuts and where to report bugs or suggest improvements.

3. Measuring a V-curve

1. Set up the bench as usual: lens under test + mirror at the flange plane; reticle visible and roughly focused.
2. Select the **G channel** (button CHANNEL or key `g`).
3. Place the ROI over the reticle image.
4. Move the lens/stage to one side of best focus. For each step:
   - Press **POINT** (or space bar).
   - Type the **micrometer position** on the on-screen pad. It opens with the **previous value selected** (shown inverted): just type to replace it, or press OK to repeat it. The `-` key toggles a negative sign; the **mm | um** toggle at the top switches the input unit (μm needs no decimal point: type `550` instead of `0.55`). Press **OK**.
5. Take **at least 5 points** (FIT unlocks at 5; 7–11 is better), bracketing the peak — go *through* best focus, don't stop at it.

6. Press **FIT**. The app saves a CSV and shows, per channel: `vertex ± sigma (points used)` and the spectral offsets `R−G` / `B−G` in µm. **Use the G vertex** as the FFD result. A **mirror tilt** line is also shown: the app tracks how the image walks sideways during the scan (the mirror rotates with the micrometer spindle, 0.5 mm/turn, so a tilted mirror traces a spiral). A large value (or a growing rms) means the mirror needs squaring.

Notes: the fit uses points at ≥60% of the peak metric; the app flushes camera buffer frames before each POINT (anti-lag), and camera auto-exposure/WB/focus are disabled automatically. Approach each position from the same direction to avoid micrometer backlash. Do not mix different cameras, ROIs or resolutions inside one scan (the app clears points when resolution changes).

4. Cameras

CAM opens the selector: each input appears as a card with a **live thumbnail**, index, resolution and name. Tap to switch. *SCAN AGAIN* re-probes; the chosen camera is remembered for the next session. On Windows, the camera's **driver panel** (exposure, gain…) is inside SETTINGS — or key `o`.

5. Settings

SETTINGS holds: the **camera resolution** (720p / 1080p default / MAX — higher sharpens focus discrimination; changing it clears the points; the top-right badge shows actual resolution and fps); the camera **DRIVER** panel (Windows only); the **photos & CSV folder** — *CHANGE* picks a new one, *OPEN* shows it in your file manager; the **zoom step** (smooth 5% — the default — 2×, or straight to MAX) and the **max zoom** (5× default, or 10×); and two gesture switches: *double-tap = MAX zoom*, and *plain scroll zooms* without Ctrl (on by default on macOS, where two-finger scrolling is the natural gesture). Everything is remembered between sessions.

6. Updates & reporting

On startup the app checks for a new version (needs internet) and only asks: **UPDATE NOW** downloads and installs it silently, then reopens the app by itself — no wizard, no clicks. **LATER** dismisses. If the automatic update is not possible (offline, Linux `.deb` install), the download page opens instead. You can also tap **CHECK FOR UPDATE** inside **?**. To report a bug or request a feature, tap **SEND REPORT** in **?**: type your message (on-screen or physical keyboard) and send — it goes straight to the support inbox. Offline, an E-MAIL button appears as fallback.

7. Files (default `Documents/colimador/`, changeable in SETTINGS)

File	Content
`vcurve_YYYYMMDD_HHMMSS_WxH.csv`	raw points: `pos_mm, tenengrad_R, G, B, cx_px, cy_px` (positions always in **mm**; the centroid columns feed the mirror-tilt fit; the filename records the camera resolution used)
`snap_*.png`	screen snapshots (PHOTO button)
`config.json`	language, unit, last camera, folder, zoom & gesture options

8. Keyboard shortcuts

`space` POINT · `f` FIT · `g r b w` channel · `c` clear points · `+ −` zoom · `o` driver (Win) · `ESC` / `q` quit · keypad: digits, `−`, `.`, Backspace, Enter = OK.

9. Troubleshooting

- **"No camera found"** — plug the USB camera and tap the screen to rescan (CLOSE button quits).
- **Black or dark image** — open DRIVER (Windows) and raise exposure; check the collimator lamp.
- **"fit failed"** — the points don't bracket a peak: add points on both sides of best focus, or widen the scan range.
- **Numbers look 1000× off** — check the mm | um toggle on the keypad.

Collimator VCurve v5.4 — 2olhares (Lucas Werneck). The CSV is always in mm. Minimum 5 points per fit. G channel is the official measurement.